Vivekananda Giri

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Summary

PhD candidate at IIT Bombay working at the intersection of computer vision and multimodal AI, focusing on vision–language models, prompt learning, and domain generalization, with emphasis on security and privacy in visual understanding.

Education

2025 – present	PhD, Computer Vision & Multimodal Deep Learning — **IIT Bombay**	GPA 8.71/10
2023 – 2025	MTech — **IIT Kharagpur**	GPA 9.36/10
2019 – 2023	BTech — **Jadavpur University**	GPA 8.76/10
2017 – 2019	Higher Secondary Education	94.8%
2017	Secondary Education	95.57%

Publications

From Patches to Pixels: Dual-Branch Prompt Learning for Hyperspectral Scene Generalization

V. Giri, U. Chaudhuri, B. Banerjee — British Machine Vision Conference (BMVC), 2026 (Accepted). [Project Page]

- Dual-branch framework unifying patch-level (CLIP) and pixel-level (SAM-2) features via a Token-wise Cross-Scale Distillation Module (TCDM).
- Prompt-Conditioned Low-Rank Adapter (PCLRA) and Hierarchical Orthogonality Regularizer (HOR) for diverse, context-aware text-encoder adaptation.
- Gated Product-of-Experts (GPoE) fusion for robust, well-calibrated predictions under spectral shift.

Work Experience

ML Research Intern — Tata Steel R&D Link Jun 2024 – Nov 2024

- Built an 8-layer Convolutional Autoencoder (from scratch) for microstructural image compression/decompression.
- Improved reconstruction via hyperparameter tuning and ResNet/DenseNet skip connections; validated with MATLAB contour-overlay analysis.

Projects

Medical-VLM (Course Project) Link
- Analyzed textual–visual alignment in medical VLMs (ALIGNVLM) for chest X-ray report generation; ViT-B/16 + DistilGPT2 on IU X-ray.

Multi-task Image Restoration (Course Project) Link
- Unified denoising + inpainting on a frozen DINOv3-Small ViT with LoRA; task-specific decoders share a latent space, coupled via cross-attention.

Few-shot Learning with MAML (Course Project) Link
- MAML with a MobileNetV3-Small backbone for 10-way 1-shot classification; cross-domain CIFAR-100 → CUB-200-2011.

Genetic Programming — Chromium Recovery (MTech Thesis) Link
- Lasso / Ridge / Random-Forest regression with GP-engineered symbolic features for chromium-recovery prediction.

Coursework

Intro to Deep Learning, Advanced Deep Learning for Computer Vision, Deep Learning for NLP, Digital Image Processing, Machine Learning, Numerical Methods.

Skills

Languages	Python, C++
Frameworks	PyTorch, Hugging Face (Transformers, Diffusers), OpenCV
Research areas	Vision–Language Models (CLIP, SAM-2, ViT), Prompt Learning (CoOp, MaPLe), LoRA / PEFT, Meta- & Few-shot Learning (MAML), Diffusion / VAE

Academic Achievements

- AIR 87 at the National Engineering Olympiad (NEO 6.0); top 500 in WBJEE; top 2 percentile in JEE Main & Advanced.
- 22nd rank (Class X) and 20th rank (Class XII).